Classical Mechanics

Lecture 9: Canonical Transformations, Generators, and Symmetry

Lectures by Leonard Susskind

Transcript-derived companion notes curated by LazyingArt LLC
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Chapter 1

Canonical Transformations, Generators, and Symmetry

Phase space is not merely a list of positions and momenta. It has a structure, and the structure is what remains meaningful when the coordinates used to describe it are changed. The Poisson bracket is the local expression of that structure. It tells us which changes of phase-space coordinates are allowed, how functions flow under those changes, and why a symmetry carries a conserved quantity with it.

1.1 Structure and Flow on Phase Space

There are many kinds of structure one can place on a space. A Riemannian metric, for example, supplies a distance between neighboring points. A Poisson structure is different: it supplies an antisymmetric bracket between functions. For mechanics the coordinates of the space are not only $q_1, \ldots, q_n$, but the full set

$$(q_1, \ldots, q_n, p_1, \ldots, p_n).$$

The question is therefore richer than a change of coordinates in ordinary space. We want to know which transformations may mix the q's and p's without destroying the form of mechanics.

It is useful to think first about familiar transformations. A finite rotation is built from many small rotations, and a finite translation from many small translations. The intermediate configurations trace a flow. The flow need not be the actual motion of a physical system; it can simply carry one description into another.

The same idea applies to phase space. Hamiltonian evolution is one flow: a phase point moves with time under the influence of H. Rotations, translations, and less familiar transformations that mix positions and momenta define other flows. Poisson brackets give one language for all of them. Newton's equations are the most concrete end of this story; the Poisson formulation is the abstract end needed here.

Ordinary mechanical systems most visibly exhibit translations and rotations, although special systems can hide additional symmetries. More advanced physics adds internal symmetries that are not motions of ordinary space at all. The common question is structural: what can be transformed while the laws retain their form? Hamilton and Poisson supplied the classical language in which

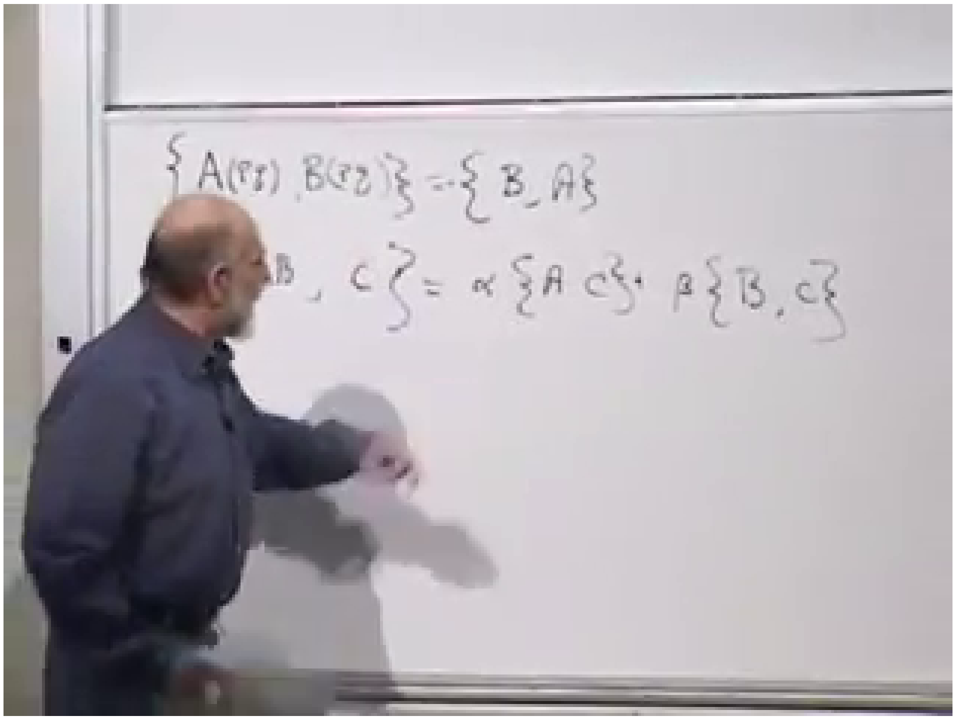

Figure 1.1: Antisymmetry and linearity on the board at 00:09:12. The minus sign in the first rule is the feature that fixes the order of every later bracket.

that question can be asked directly on phase space.

1.2 The Poisson Algebra

Let A, B, and C be functions on phase space. The first defining property is antisymmetry,

$$\{A, B\} = -\{B, A\}. \tag{1.1}$$

The second is linearity in each entry. For ordinary numbers α and β,

$$\{\alpha A + \beta B, C\} = \alpha\{A, C\} + \beta\{B, C\}. \tag{1.2}$$

Question from the audience. Should the first relation have a minus sign when A and B are interchanged?[a]

Response. Yes. That minus sign is the essential point: $\{A, B\} = -\{B, A\}$. Without it there would be no distinction between the two orders.

[a] Lecture timestamp: 00:08:09–00:08:18.

The bracket also obeys the product rule,

$$\{AB, C\} = A\{B, C\} + B\{A, C\}. \tag{1.3}$$

By antisymmetry the corresponding rule in the second entry follows as well.

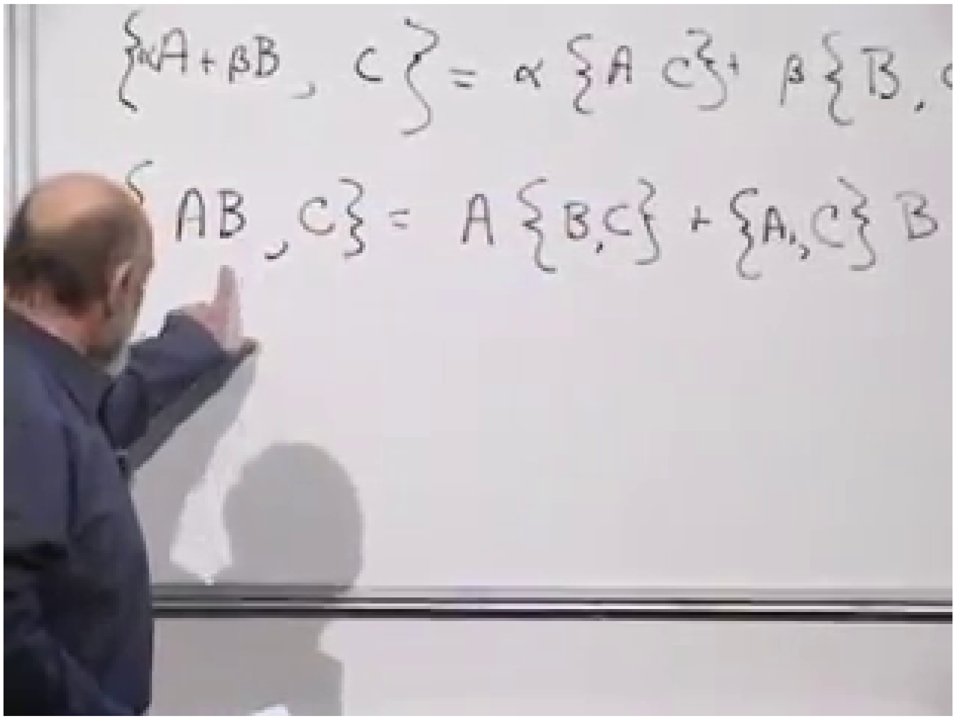

Figure 1.2: The product rule at 00:11:18. In classical mechanics the functions outside the bracket multiply as ordinary commuting numbers.

The order of multiplication outside a bracket is immaterial: $A\{B, C\} = \{B, C\}A$. The order *inside* the bracket is not immaterial, because reversing it changes the sign. This distinction is easy to overlook in classical mechanics, where functions commute, but it anticipates the ordering questions that become unavoidable in quantum mechanics.

Question from the audience. What sort of objects are A and B? Must they be real-valued scalars?[a]

Response. They are numerical-valued functions of all the phase-space coordinates. They may be added and multiplied pointwise, and complex-valued functions are allowed when useful. Here *numerical* only means that their ordinary multiplication is commutative; it is separate from the physicist's use of *scalar* to mean invariant under spatial rotations.

[a]Lecture timestamp: 00:09:26–00:14:13.

For canonical coordinates the concrete derivative formula is

$$\{A, B\} = \sum_{i=1}^{n} \left(\frac{\partial A}{\partial q_i} \frac{\partial B}{\partial p_i} - \frac{\partial A}{\partial p_i} \frac{\partial B}{\partial q_i} \right). \tag{1.4}$$

We assume throughout that the functions are smooth enough for the indicated derivatives and limiting arguments.

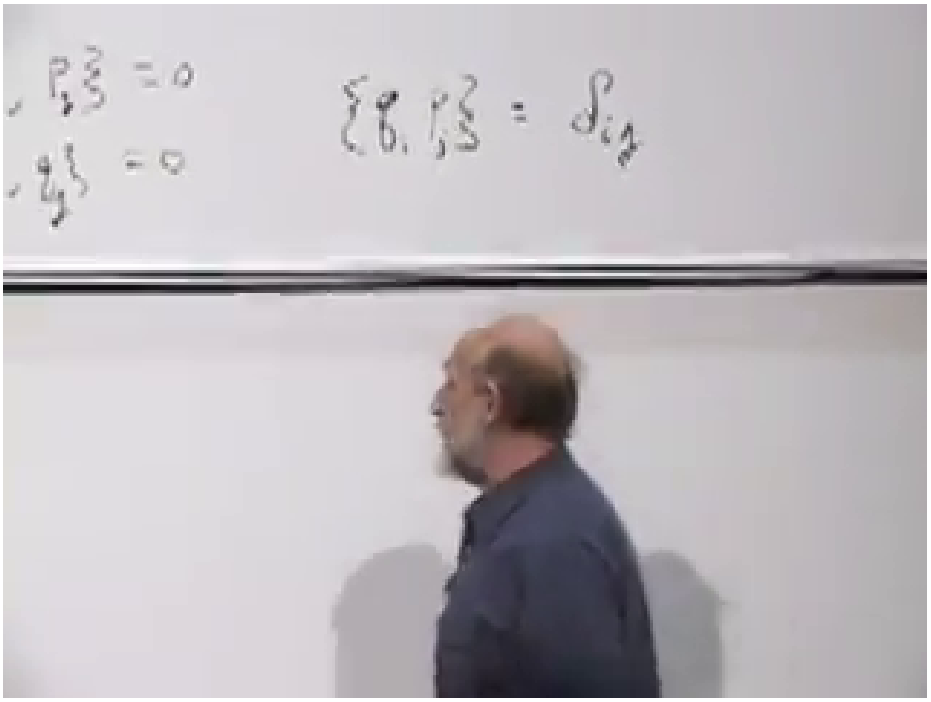

Figure 1.3: The canonical brackets at 00:17:25. The mixed bracket is the Kronecker delta; brackets among two coordinates or two momenta vanish.

1.3 Canonical Brackets as Derivatives

Antisymmetry immediately gives every self-bracket:

$$\{p_i, p_i\} = 0, \qquad \{q_i, q_i\} = 0.$$

The derivative definition gives the complete canonical relations,

$$\{q_i, q_j\} = 0, \qquad \{p_i, p_j\} = 0, \qquad \{q_i, p_j\} = \delta_{ij}. \tag{1.5}$$

Consequently $\{p_i, q_j\} = -\delta_{ij}$. The Kronecker delta is one when $i = j$ and zero otherwise.

Antisymmetry determines a self-bracket because it equals its own negative. For two different variables it says only $\{q, p\} = -\{p, q\}$; the normalization $\{q, p\} = 1$ comes from the canonical derivative definition.

Two useful zeroes follow at once:

$$\{F(q), K(q)\} = 0, \qquad \{G(p), L(p)\} = 0. \tag{1.6}$$

Each term in a Poisson bracket contains one derivative with respect to a coordinate and one with respect to a momentum. A nonzero contribution therefore needs a coordinate dependence on one side to meet the conjugate momentum dependence on the other.

The fundamental brackets and the product rule turn bracketing into differentiation. For one canonical pair,

$$\{F(q), p\} = \frac{dF}{dq}. \tag{1.7}$$

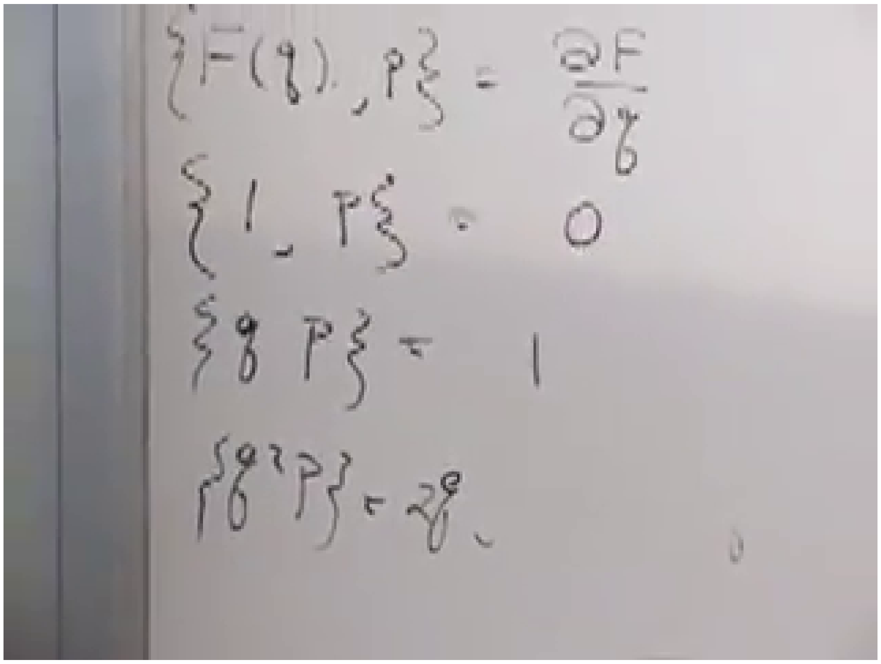

Figure 1.4: The monomial proof at 00:22:43: the cases 1, q, and q^2 reproduce their ordinary derivatives.

Rather than simply reading this from Eq. (1.4), let us recover it from the algebra.

For the constant function,

$$\{1^2, p\} = 1\{1, p\} + \{1, p\}1 = 2\{1, p\}. \tag{1.8}$$

Since $1^2 = 1$, this says $\{1, p\} = 2\{1, p\}$, hence $\{1, p\} = 0$. Next,

$$\{q, p\} = 1.$$

For the next monomial,

$$\{q^2, p\} = q\{q, p\} + \{q, p\}q = 2q. \tag{1.9}$$

Induction gives $\{q^r, p\} = rq^{r-1}$. Linearity extends the result to polynomials and, under the smoothness assumption, to the functions used in mechanics. In many variables the paired identities hold for a general phase-space function:

$$\{F(\boldsymbol{q}, \boldsymbol{p}), p_i\} = \frac{\partial F}{\partial q_i}, \qquad \{q_i, F(\boldsymbol{q}, \boldsymbol{p})\} = \frac{\partial F}{\partial p_i}. \tag{1.10}$$

The order is deliberate: p_i goes on the right to differentiate with respect to q_i, whereas q_i goes on the left to differentiate with respect to p_i.

Question from the audience. Does the extension from polynomials require continuity of the bracket and of the functions?[a]

Response. Yes. We are assuming smooth, multiply differentiable functions and the usual continuity needed to pass from polynomial approximations to their limits.

[a]Lecture timestamp: 00:24:11–00:24:38.

1.4 Hamiltonian Evolution Is One Flow

Dynamics enters through one further relation:

$$\dot{A} = \{A, H\}. \tag{1.11}$$

A phase point follows an orbit determined by H. As the point moves, a function $A(q, p)$ changes along that orbit, and Eq. (1.11) is its directional derivative along the Hamiltonian flow.

Question from the audience. From the abstract viewpoint, are the p's and q's being assumed to depend on time?[a]

Response. Apply the same rule to the coordinates themselves: $\dot{q}_i = \{q_i, H\}$ and $\dot{p}_i = \{p_i, H\}$. Their time dependence is then defined by the Hamiltonian flow.

[a]Lecture timestamp: 00:27:20–00:27:52.

For a free particle in one dimension,

$$H = \frac{p^2}{2m}. \tag{1.12}$$

The momentum equation is

$$\dot{p} = \left\{p, \frac{p^2}{2m}\right\} = 0, \tag{1.13}$$

because both entries depend only on p. The coordinate equation is

$$\begin{aligned}\dot{q} &= \left\{q, \frac{p^2}{2m}\right\} \\ &= \frac{1}{2m}\left(p\{q, p\} + \{q, p\}p\right) = \frac{p}{m}.\end{aligned} \tag{1.14}$$

Thus the abstract bracket algebra reproduces Hamilton's equations and the ordinary velocity–momentum relation.

Question from the audience. Is this Poisson formulation restricted to nonrelativistic mechanics?[a]

Response. No. The example $H = p^2/(2m)$ is nonrelativistic, but the canonical structure applies much more broadly: relativistic particles, classical fields, electromagnetism, and general relativity all admit the appropriate Hamiltonian form. Quantum mechanics replaces the classical bracket by its operator counterpart.

[a]Lecture timestamp: 00:29:57–00:30:45.

1.5 Which Coordinate Changes Preserve Mechanics?

We now separate two questions. First, which changes of phase-space coordinates preserve the Poisson structure? Second, among those changes, which leave a particular Hamiltonian invariant? The first question defines a canonical transformation; the second will define a symmetry.

Take one canonical pair (q, p). A simultaneous stretch,

$$Q = 2q, \qquad P = 2p, \tag{1.15}$$

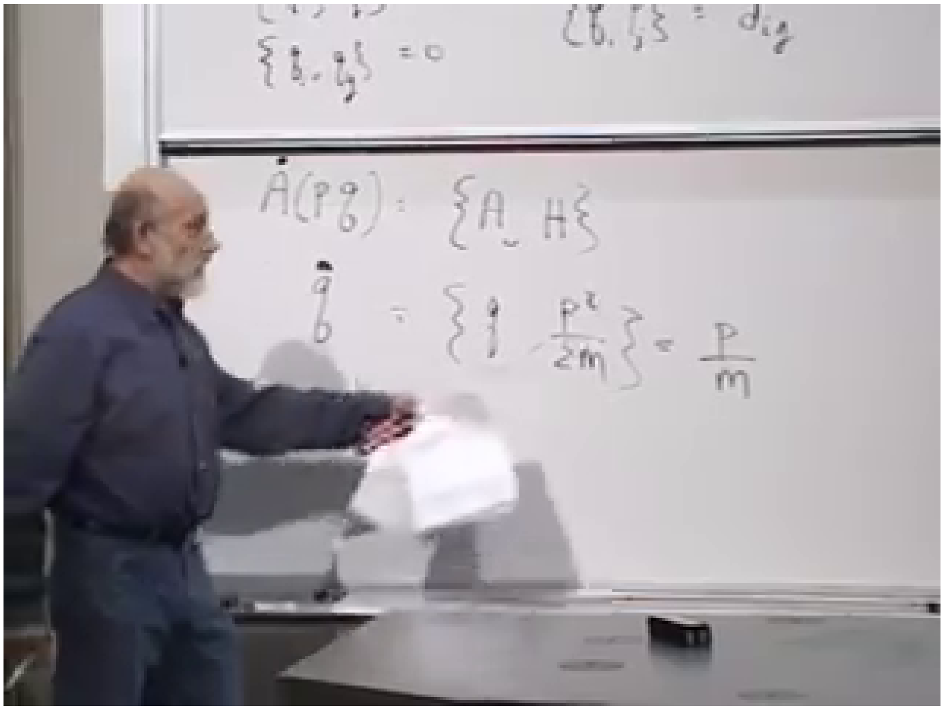

Figure 1.5: The free-particle calculation at 00:29:22. The Poisson algebra gives $\dot{p} = 0$ and $\dot{q} = p/m$ directly.

fails because

$$\{Q, P\} = 4\{q, p\} = 4.$$

A compensated stretch and squeeze,

$$Q = 2q, \qquad P = \frac{1}{2}p, \tag{1.16}$$

succeeds:

$$\{Q, P\} = \left\{2q, \frac{1}{2}p\right\} = 1.$$

For one pair, the successful transformation preserves area in the (q, p)-plane. The double stretch does not. In many dimensions, preserving total volume is not enough: the transformation must preserve the full symplectic form

$$\omega = \sum_i dq_i \wedge dp_i,$$

or equivalently the complete canonical bracket relations.

Question from the audience. Is the squeeze-and-stretch condition basically conservation of area?[a]

Response. For one q, p pair, yes: canonical transformations are area-preserving maps of that plane. With several pairs, ordinary volume preservation is only a consequence; preservation of the full symplectic structure is the stronger condition.

[a] Lecture timestamp: 00:34:51–00:35:20.

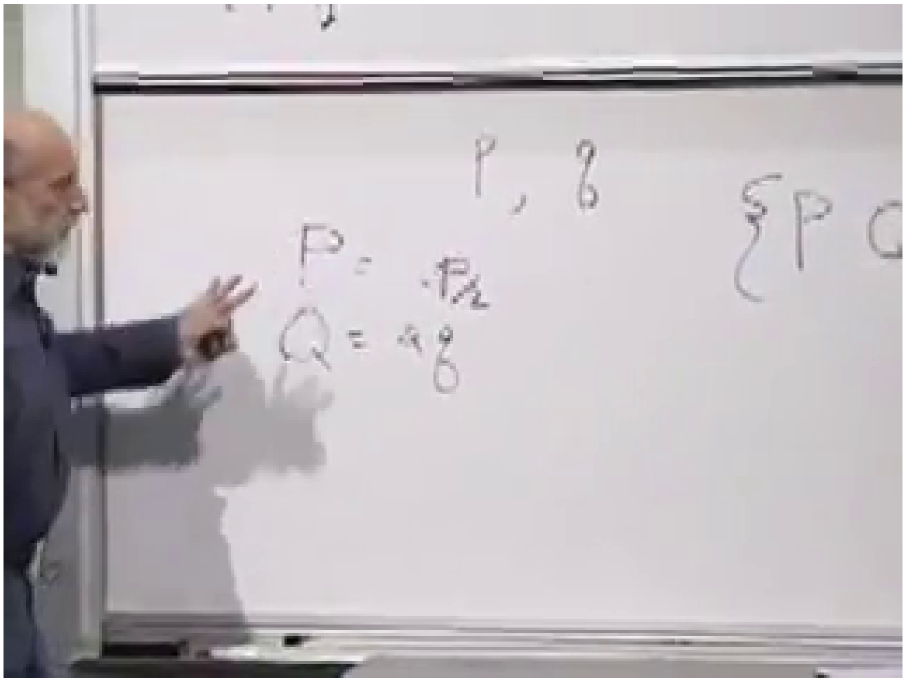

Figure 1.6: The compensated rescaling at 00:33:55. Stretching q while squeezing p preserves the canonical bracket.

A rotation of the phase-space plane also works:

$$P = \cos\theta\, p + \sin\theta\, q, \qquad Q = -\sin\theta\, p + \cos\theta\, q. \tag{1.17}$$

Self-brackets vanish by antisymmetry, so only the mixed bracket needs work:

$$\begin{aligned}\{Q, P\} &= \{-\sin\theta\, p + \cos\theta\, q,\, \cos\theta\, p + \sin\theta\, q\}\\ &= \cos^2\theta\,\{q, p\} - \sin^2\theta\,\{p, q\}\\ &= \cos^2\theta + \sin^2\theta = 1.\end{aligned} \tag{1.18}$$

Definition 1.1. A transformation

$$(q_i, p_i) \longmapsto (Q_i, P_i)$$

is canonical when

$$\{Q_i, Q_j\} = 0, \qquad \{P_i, P_j\} = 0, \qquad \{Q_i, P_j\} = \delta_{ij}.$$

Question from the audience. Are the equations of motion unchanged after a canonical transformation?[a]

Response. Their canonical form is unchanged, but the Hamiltonian must be re-expressed as a function of the new variables Q and P. The numerical formula for H need not look the same.

[a]Lecture timestamp: 00:40:04–00:40:20.

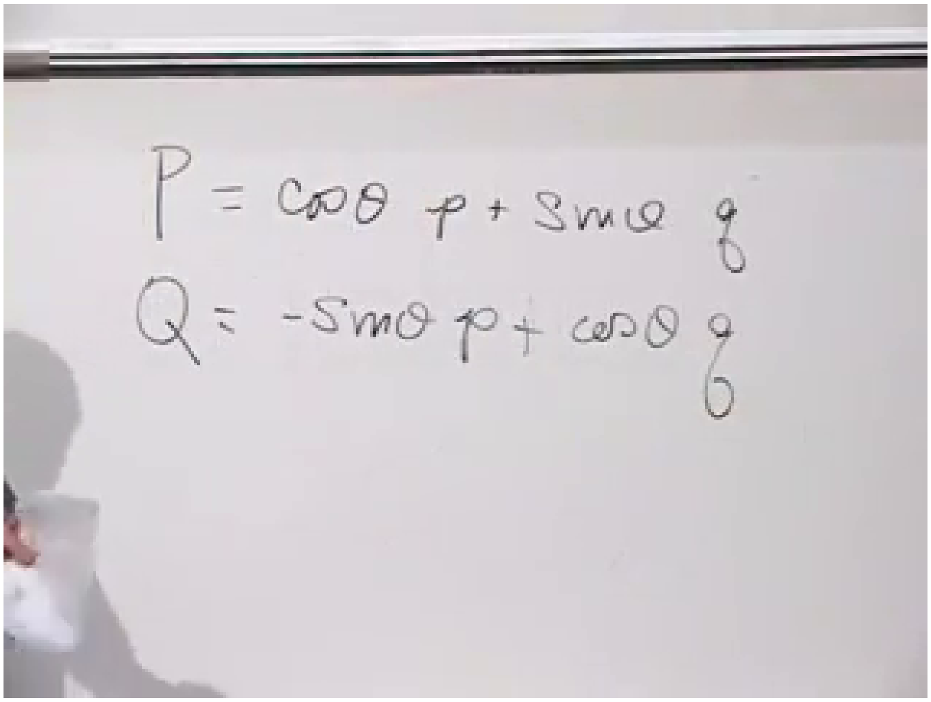

Figure 1.7: A phase-space rotation at 00:36:10. Positions and momenta are mixed, but the canonical structure survives.

Question from the audience. Are canonical transformations examples of gauge transformations?[a]

Response. No. A gauge transformation relates redundant descriptions of the same physical state. A canonical transformation is an ordinary change of coordinates on phase space that preserves its Poisson structure.

[a]Lecture timestamp: 00:40:21–00:41:05.

Question from the audience. Can a canonical transformation carry one arbitrary physical system into another?[a]

Response. Here it means a change of phase-space coordinates for the same system. Phase spaces with the same number of degrees of freedom may be locally isomorphic, but a canonical re-description does not by itself turn a harmonic oscillator into an unrelated many-body system.

[a]Lecture timestamp: 00:41:42–00:44:52.

1.6 Infinitesimal Transformations and Their Generators

Question from the audience. Is there a more constructive test than checking all the new Poisson brackets one by one?[a]

Response. For an infinitesimal transformation there is. Build the change from a generator $G(q, p)$. Finite transformations connected to the identity can then be assembled from these small steps.

[a]Lecture timestamp: 00:45:07–00:46:31.

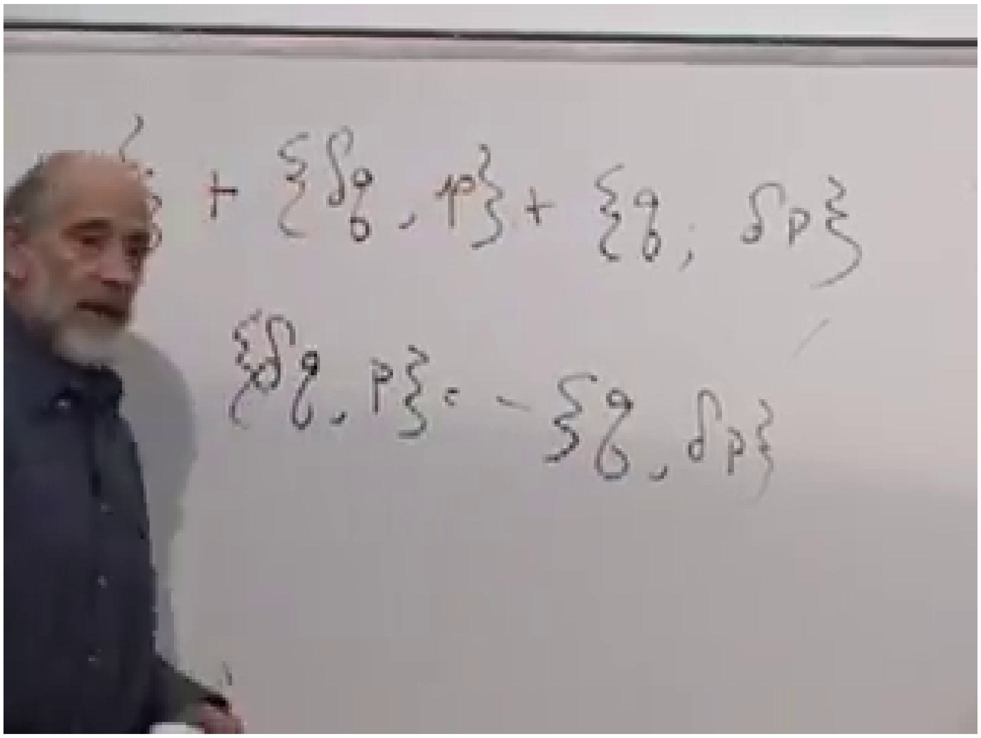

Figure 1.8: The first-order canonical condition at 00:50:44: $\{\delta q, p\} = -\{q, \delta p\}$.

Write one small step as

$$Q = q + \delta q, \qquad P = p + \delta p, \tag{1.19}$$

where δq and δp are themselves functions of phase space. Infinitesimal means that terms quadratic and higher in these changes may be discarded. Expanding the transformed bracket gives

$$\begin{aligned} \{Q, P\} &= \{q + \delta q, p + \delta p\} \\ &= \{q, p\} + \{\delta q, p\} + \{q, \delta p\} + \{\delta q, \delta p\}. \end{aligned} \tag{1.20}$$

The last term is second order. Therefore the first-order canonical condition is

$$\{\delta q, p\} = -\{q, \delta p\}. \tag{1.21}$$

Introduce a small parameter ϵ and a function $G(q, p)$. Define

$$\delta q = \epsilon\{q, G\}, \qquad \delta p = \epsilon\{p, G\}. \tag{1.22}$$

Using Eq. (1.10),

$$\delta q = \epsilon \frac{\partial G}{\partial p}, \qquad \delta p = -\epsilon \frac{\partial G}{\partial q}. \tag{1.23}$$

These formulas solve the canonical condition automatically:

$$\begin{aligned} \{\delta q, p\} &= \epsilon \left\{ \frac{\partial G}{\partial p}, p \right\} = \epsilon \frac{\partial^2 G}{\partial q\, \partial p}, \\ -\{q, \delta p\} &= \epsilon \left\{ q, \frac{\partial G}{\partial q} \right\} = \epsilon \frac{\partial^2 G}{\partial p\, \partial q}. \end{aligned} \tag{1.24}$$

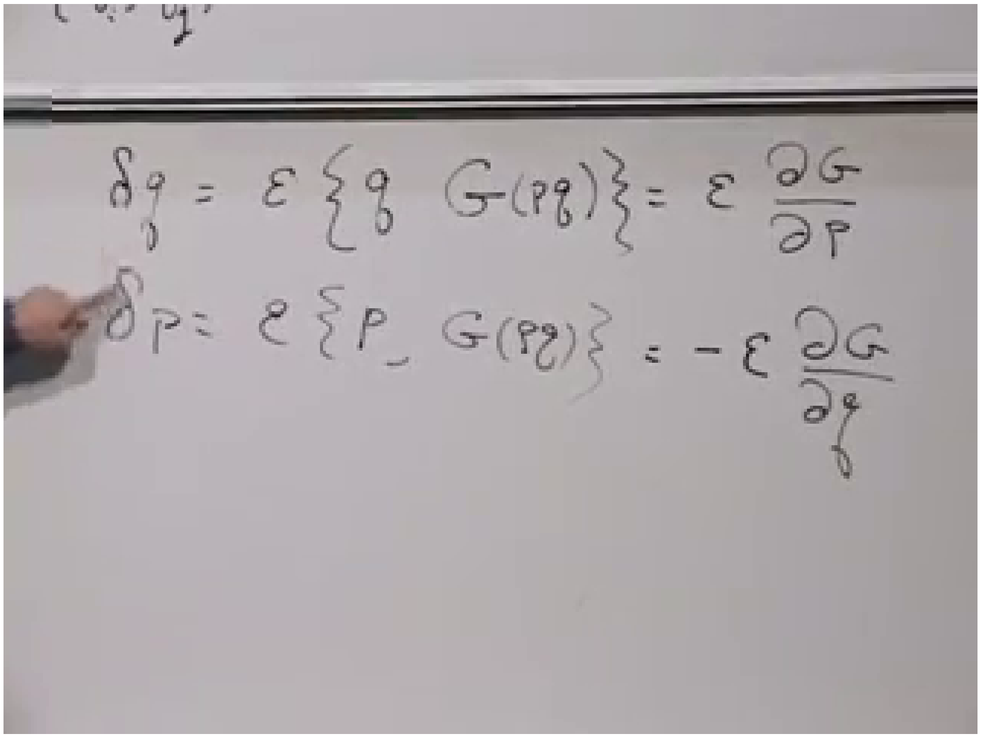

Figure 1.9: The generator equations at 00:53:55. A single function G determines both components of the infinitesimal phase-space flow.

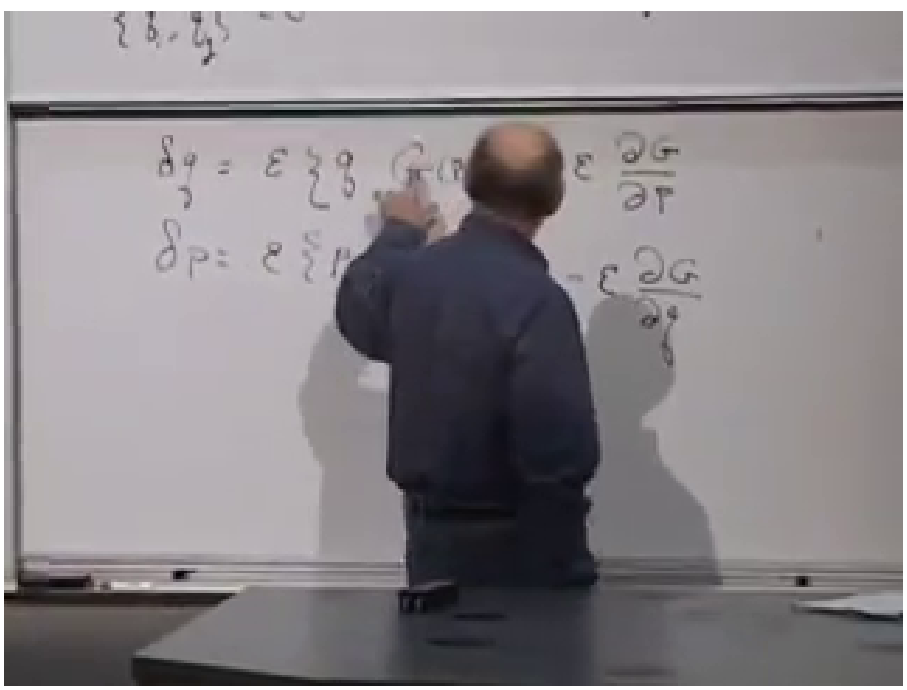

Figure 1.10: The mixed-derivative check at 00:55:50. It proves that a generator-defined infinitesimal flow is canonical.

The equality is simply equality of mixed partial derivatives.

Question from the audience. Must the two variations use the bracket with G in the same order?[a]

Response. Yes: $\delta q = \epsilon\{q, G\}$ and $\delta p = \epsilon\{p, G\}$. The opposite signs in their derivative forms come from the canonical bracket itself, not from reversing the order.

[a]Lecture timestamp: 00:52:08–00:52:15.

The same generator controls every function. By the chain rule,

$$\begin{aligned}\delta A &= \frac{\partial A}{\partial q}\delta q + \frac{\partial A}{\partial p}\delta p \\ &= \epsilon\left(\frac{\partial A}{\partial q}\frac{\partial G}{\partial p} - \frac{\partial A}{\partial p}\frac{\partial G}{\partial q}\right) \\ &= \epsilon\{A, G\}. \end{aligned} \tag{1.25}$$

For many degrees of freedom the sum over i is understood, and each δq_i may depend on every q_j and p_j, not only on the pair with the same index.

Hamiltonian evolution is the special choice

$$G = H, \qquad \epsilon = \delta t.$$

Thus time evolution is itself a canonical transformation. More generally, finite canonical transformations in the component connected to the identity are built by composing infinitesimal generator flows. This is the precise local statement needed here; global topology can introduce qualifications.

Question from the audience. If G generates a flow in the same way as H, is it just the Hamiltonian of another system?[a]

Response. Mathematically it may be treated that way. Any suitable G defines a Hamiltonian-like flow. Which function is the actual energy of the physical system is additional physical information, fixed by the system and ultimately by experiment.

[a]Lecture timestamp: 00:58:35–00:59:14.

1.7 Symmetry and Conservation

A canonical transformation preserves the form of mechanics. A symmetry of a particular system must do more: it must preserve that system's Hamiltonian. Suppose H generates the physical time flow and G generates another canonical flow. In general the G-flow crosses surfaces of constant energy. It is a symmetry precisely when it runs along them.

Equation (1.25) applied to $A = H$ gives

$$\delta H = \epsilon\{H, G\}. \tag{1.26}$$

The symmetry condition is therefore

$$\{H, G\} = 0. \tag{1.27}$$

By antisymmetry, $\{G, H\} = 0$. If G has no explicit time dependence, Hamiltonian evolution gives

$$\dot{G} = \{G, H\} = 0. \tag{1.28}$$

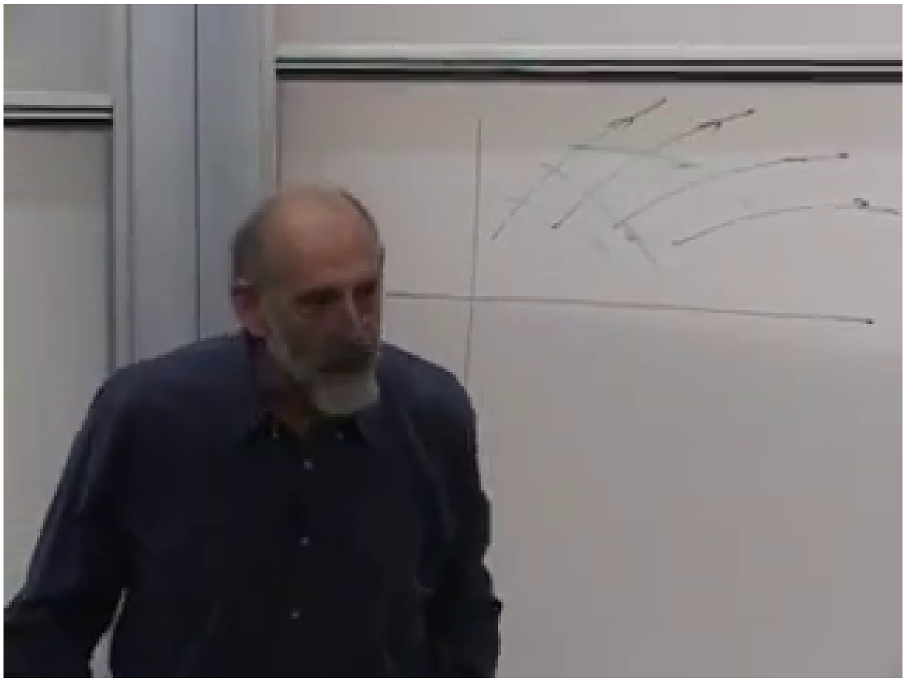

Figure 1.11: The phase-space flow picture at 01:02:10. The curved trajectories represent Hamiltonian motion; a second flow is a symmetry when it remains on the same energy surface.

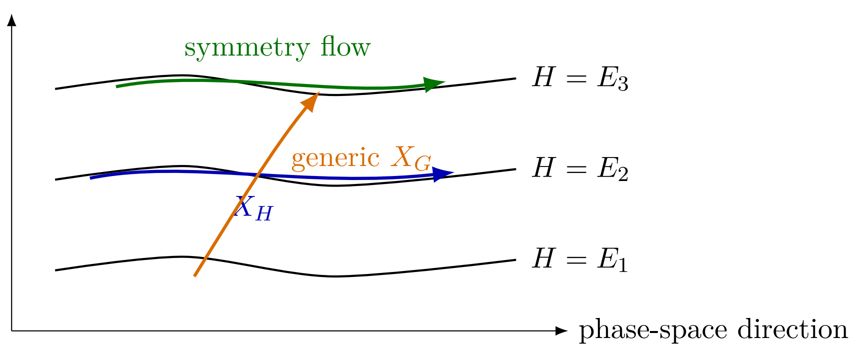

Figure 1.12: Reconstruction of the board geometry: a generic generator crosses energy surfaces, whereas a symmetry generator is tangent to them.

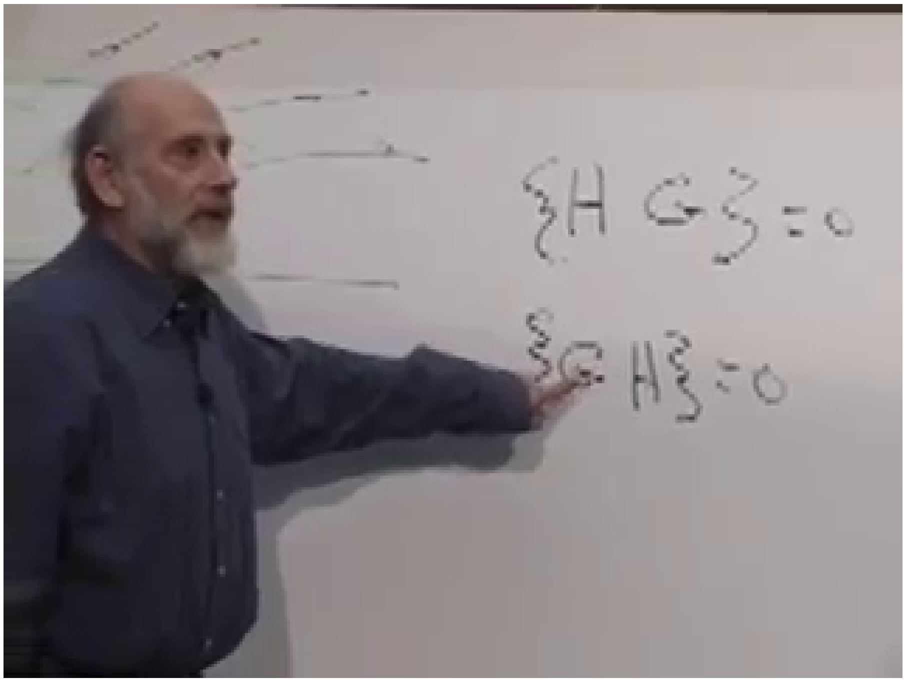

Figure 1.13: The symmetry condition at 01:05:44: $\{H, G\} = 0$ is equivalently $\{G, H\} = 0$.

This is the reciprocal core of Noether's theorem in Hamiltonian language. The G-flow does not change H, and the H-flow does not change G. A generator of a continuous symmetry is a conserved quantity.

The first statement, that G leaves the energy unchanged, is $\{H, G\} = 0$. Antisymmetry gives $\{G, H\} = 0$, and the latter bracket is precisely $\dot{G}$. They are the same vanishing bracket read along the two flows.

Question from the audience. What is a concrete example of this abstract relation? [a]

Response. Rotational symmetry and angular momentum. For a rotationally invariant Hamiltonian, the angular-momentum generator has zero Poisson bracket with H, so angular momentum is conserved.

[a] Lecture timestamp: 01:07:56–01:08:40.

1.8 Angular Momentum and the Free Particle

Take a free particle in two dimensions:

$$H = \frac{p_x^2 + p_y^2}{2m}, \qquad G = L_z = xp_y - yp_x. \tag{1.29}$$

Only canonical pairs contribute:

$$\{x, p_x\} = 1, \qquad \{y, p_y\} = 1,$$

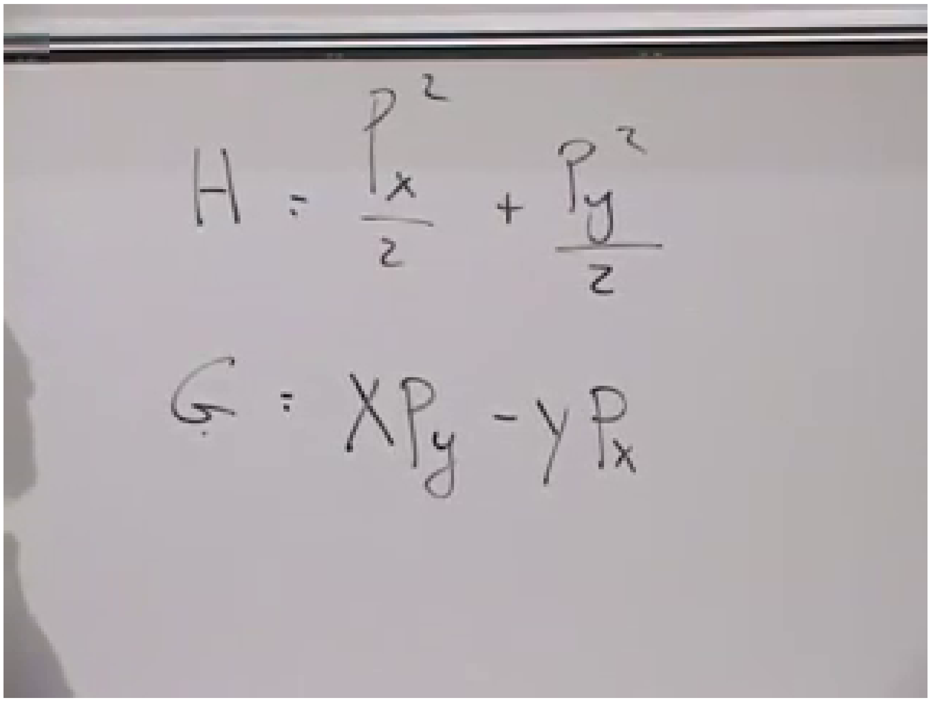

Figure 1.14: The free-particle Hamiltonian and angular-momentum generator at 01:09:35. The board sets $m = 1$; the derivation here retains m.

while $\{x, p_y\} = \{y, p_x\} = 0$. Expanding,

$$\begin{aligned}\{G, H\} &= \frac{1}{2m}\{xp_y - yp_x, p_x^2 + p_y^2\} \\ &= \frac{1}{2m}\Big(\{xp_y, p_x^2\} + \{xp_y, p_y^2\} - \{yp_x, p_x^2\} - \{yp_x, p_y^2\}\Big).\end{aligned} \tag{1.30}$$

The two cross-mismatched terms vanish. The remaining terms are

$$\begin{aligned}\{xp_y, p_x^2\} &= p_y\{x, p_x^2\} = 2p_xp_y, \\ \{yp_x, p_y^2\} &= p_x\{y, p_y^2\} = 2p_xp_y.\end{aligned} \tag{1.31}$$

They cancel:

$$\{G, H\} = \frac{1}{2m}\left(2p_xp_y - 2p_xp_y\right) = 0. \tag{1.32}$$

The same pair can be read in reverse. Invent a system whose Hamiltonian is the rotation generator,

$$H_{\mathrm{rot}} = xp_y - yp_x. \tag{1.33}$$

Question from the audience. What would it physically mean to use the generator $xp_y - yp_x$ itself as a Hamiltonian? [a]

Response. Work out Hamilton's equations. The generator then produces uniform rotation: the position and momentum vectors each move around a circle.

[a]Lecture timestamp: 01:14:30–01:16:35.

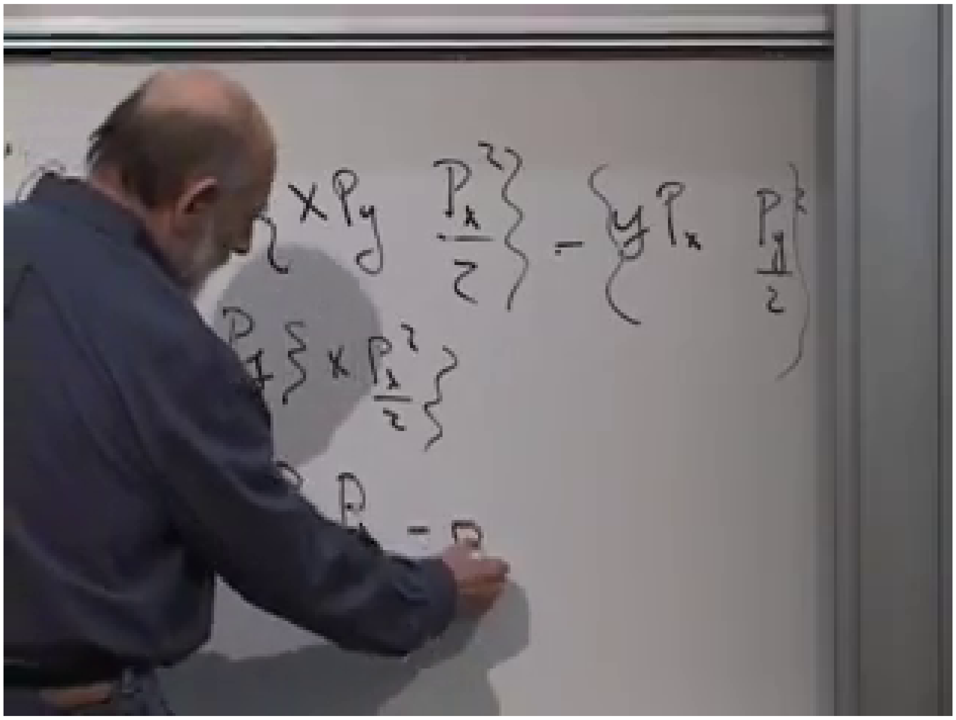

Figure 1.15: The cancellation at 01:12:50. The two surviving product-rule terms are equal and enter with opposite signs.

Hamilton's equations give

$$\dot{x} = -y, \qquad \dot{y} = x, \qquad \dot{p}_x = -p_y, \qquad \dot{p}_y = p_x. \tag{1.34}$$

Both the position vector and the momentum vector rotate uniformly. In particular,

$$\ddot{x} = -x, \qquad \ddot{y} = -y,$$

so the configuration-space orbit is a circle.

Because $\{p_x^2 + p_y^2, H_{\text{rot}}\} = 0$, the squared momentum is conserved in this invented system. The algebra does not privilege either member of a vanishing-bracket pair; physics tells us which one is the actual Hamiltonian.

Question from the audience. With many degrees of freedom, does each transformed coordinate depend only on the old pair carrying the same index? [a]

Response. No. There is one transformed Q_i and P_i for each index, but each may be a function of all the old q_j's and p_j's. The full set of canonical bracket relations constrains those coupled functions.

[a]Lecture timestamp: 01:17:01–01:18:09.

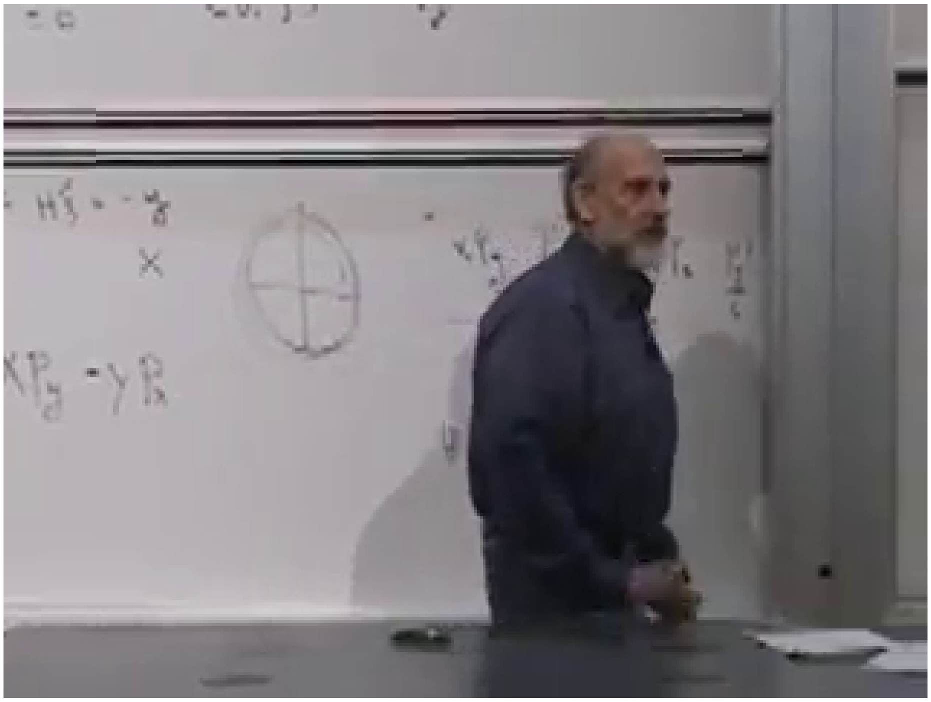

Figure 1.16: The circular orbit drawn at 01:16:35 after choosing angular momentum itself as the Hamiltonian.

Question from the audience. How is the momentum related to the velocity in Hamiltonian language? It is not always $m\dot{q}$. [a]

Response. Use the first Hamilton equation, $\dot{q}_i = \partial H/\partial p_i$, and solve it for the momenta when that relation is invertible. For $H = p^2/(2m) + U(q)$, it gives $p = m\dot{q}$. The second equation, $\dot{p} = -\partial U/\partial q$, then gives Newton's force law. More general Hamiltonians produce more general momentum–velocity relations.

[a] Lecture timestamp: 01:18:12–01:21:08.

The bracket that was introduced as an abstract algebra has now done three jobs. It generated time evolution, characterized the coordinate changes that preserve mechanics, and converted symmetry into conservation. In quantum mechanics the corresponding algebra becomes even more concrete: vanishing Poisson brackets give way to commuting operators, and generators act directly on states and observables. The classical structure already contains the map.